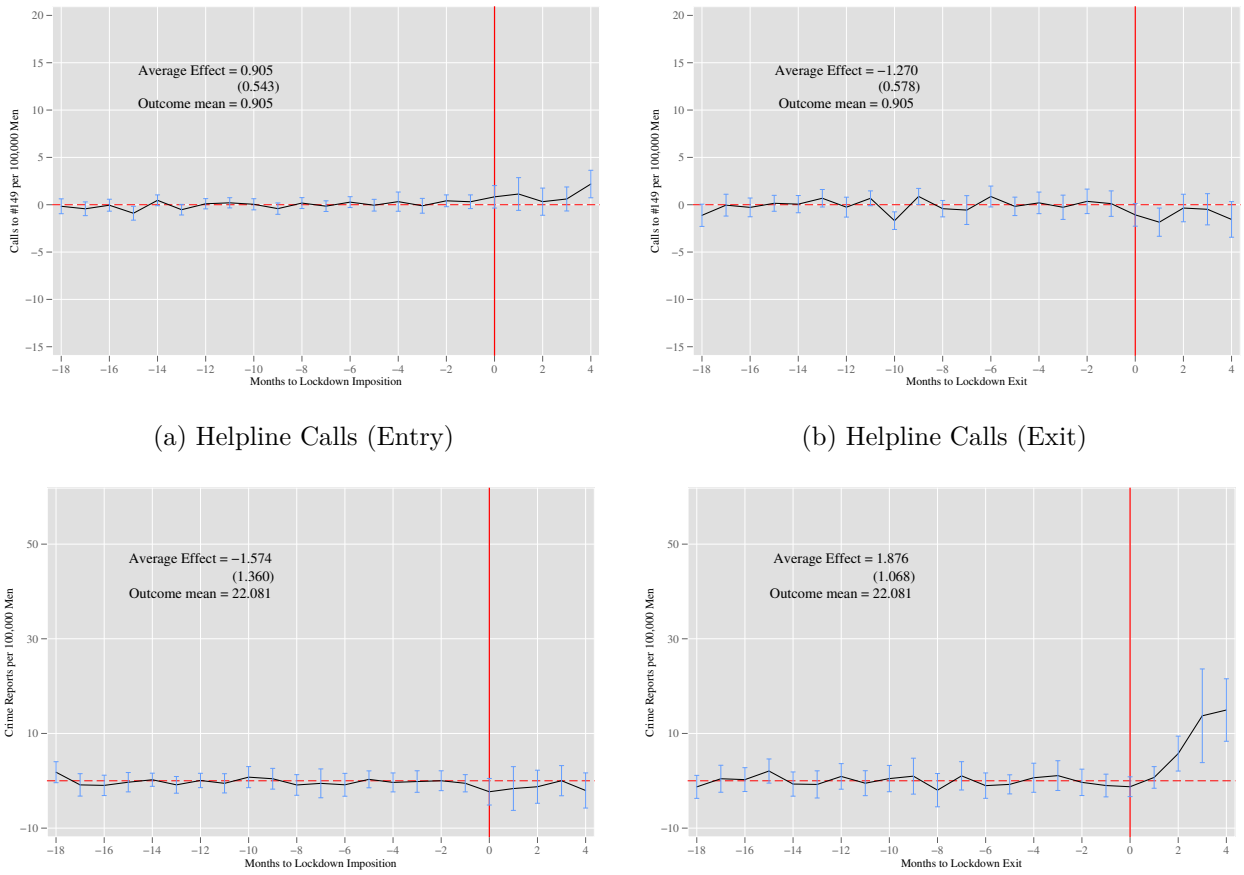
Supplementary Materials
Dynamic Impacts of Lockdown on Domestic Violence: Evidence from Multiple Policy Shifts in Chile

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A Appendix Figures and Tables

Figure A1: Impacts of Lockdown on Incidence of Intra-family Violence (Men)



(a) Helpline Calls (Entry)

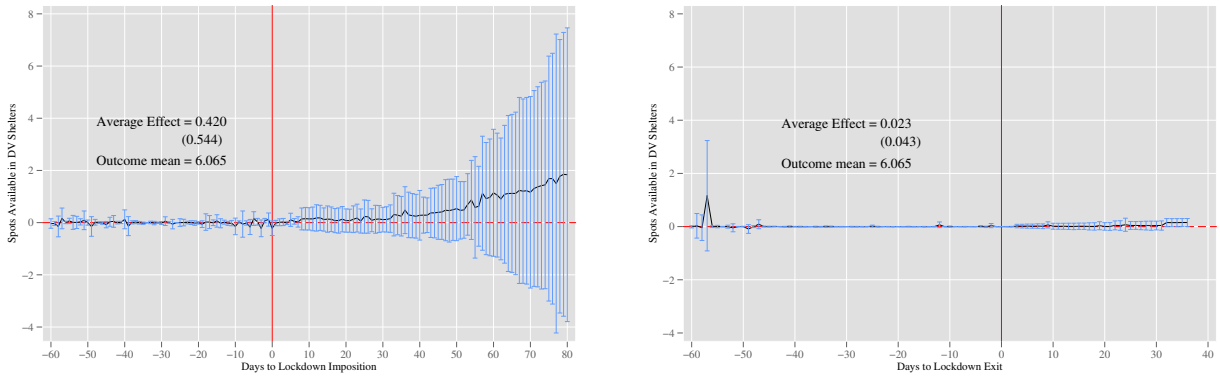
(b) Helpline Calls (Exit)

(c) Crime Reporting (Entry)

(d) Crime Reporting (Exit)

Notes: Refer to notes to Figure 2. Identical specifications are estimated however examining rates of calls to the Chilean police family support hotline (panels (a) and (b)), and formal criminal complaints (panels (c) and (d)). Calls are limited to those made by men and crimes are limited to male victims. All other estimation details follow those indicated in notes to Figure 2.

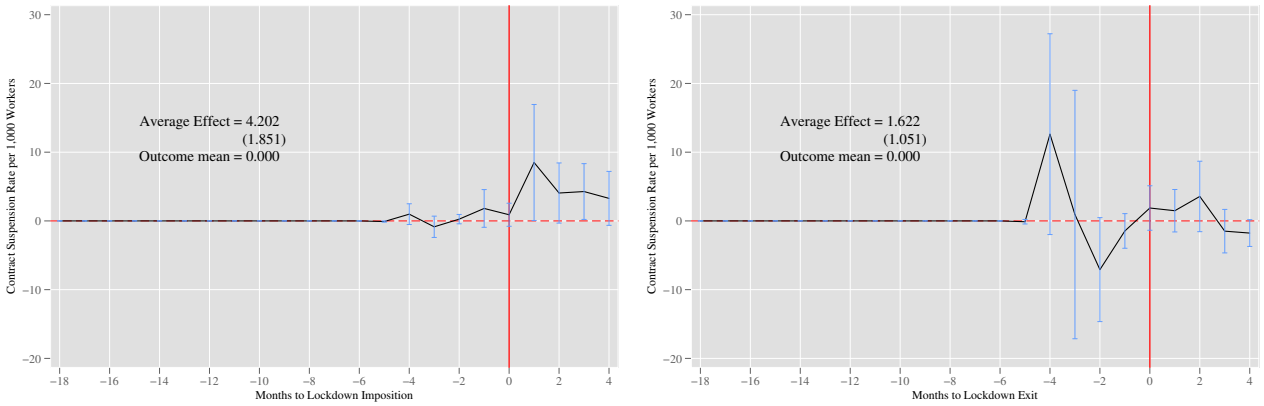
Figure A2: Lockdown Imposition and Removal and Supported Spots at DV Shelters



(a) Lockdown Entry (b) Lockdown Exit

Notes: Refer to notes to Figure 2. Identical specifications are estimated however examining available spots at DV shelters. All other estimation details follow those indicated in notes to Figure 2.

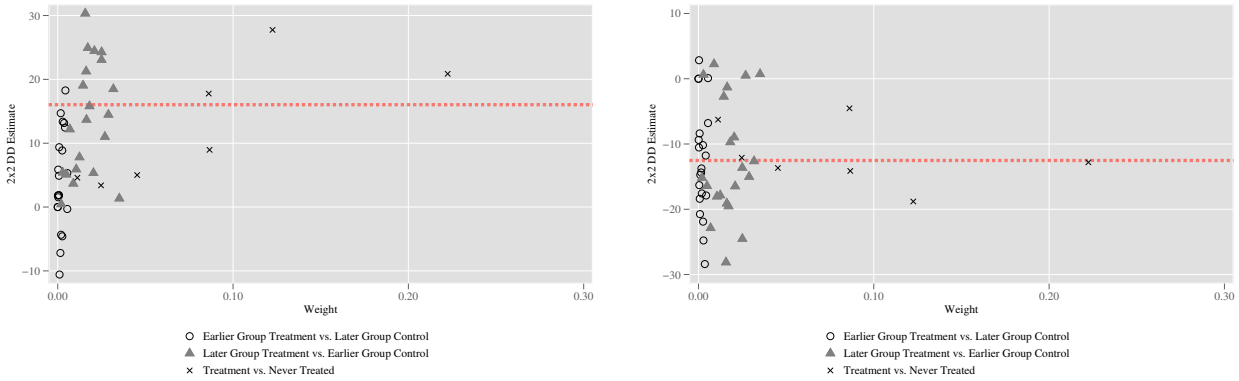
Figure A3: Lockdown Imposition and Removal and Job Suspensions



(a) Lockdown Entry (b) Lockdown Exit

Notes: Refer to notes to Figure 2. Identical specifications are estimated however examining contract suspensions. All other estimation details follow those indicated in notes to Figure 2.

Figure A4: Decomposing the Two-way Fixed Effect Estimate: Municipal-level Lockdown Entry

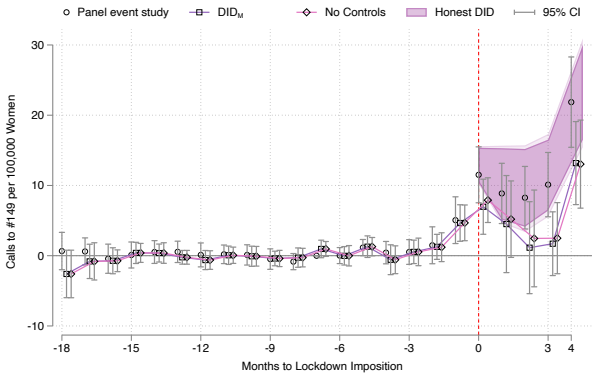


(a) Helpline Calls

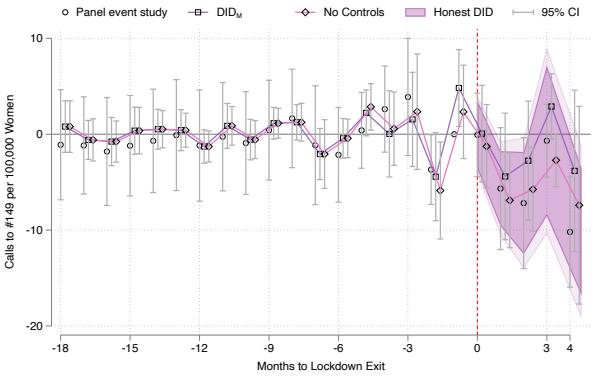
(b) Crime Reporting

Notes: Decomposition of the two-way fixed effect estimator follows Goodman-Bacon [2021], plotting each group×time estimate as well as its weight in two-way fixed effect models. Results are presented for municipal by month records, in which each of the three Goodman-Bacon [2021] groups exist (treated vs never treated, early versus late adoption, late versus early adoption). The group-specific estimates are presented as points on the plot, while the aggregate two-way FE estimate is presented as the dashed horizontal line.

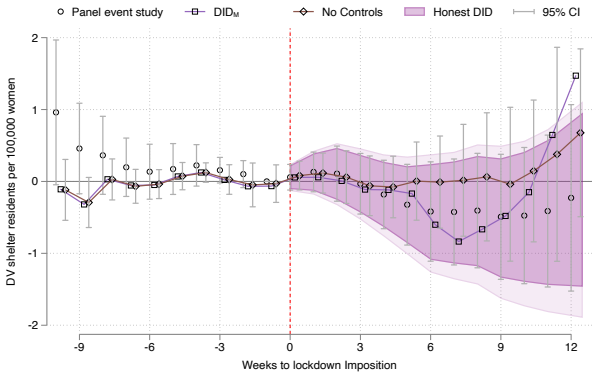
Figure A5: Estimated Impacts of Lockdown on Incidence of Intra-family Violence



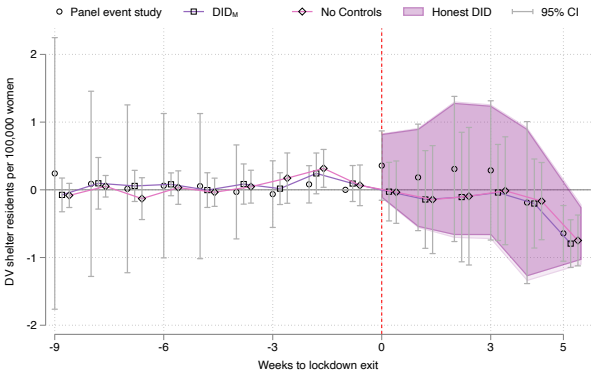
(a) Helpline Calls (Entry)



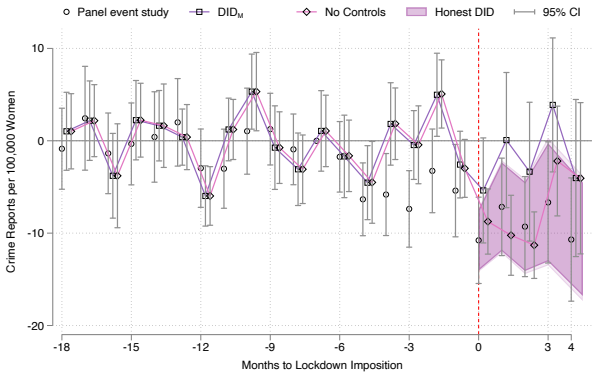
(b) Helpline Calls (Exit)



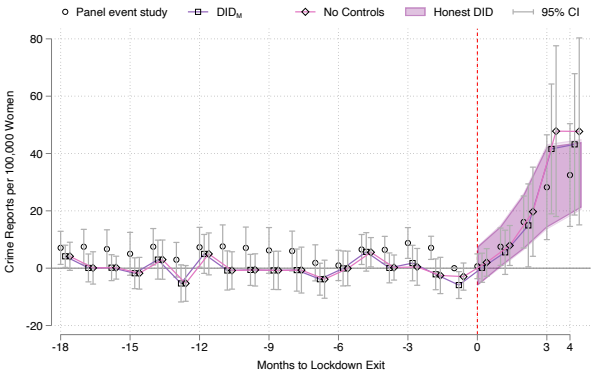
(c) Shelter Usage (Entry)



(d) Shelter Usage (Exit)



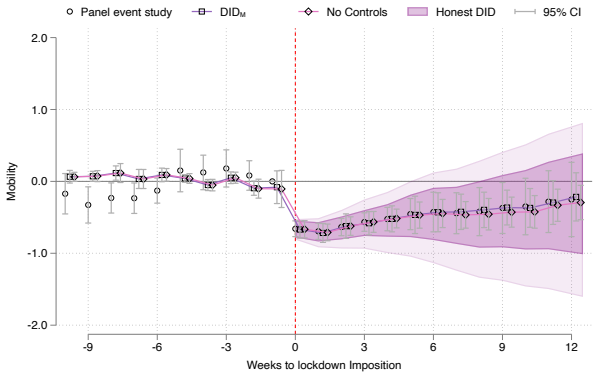
(e) Crime Reporting (Entry)



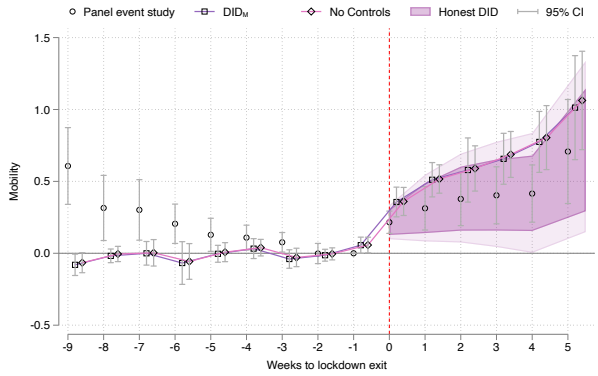
(f) Crime Reporting (Exit)

Notes: Figures display point estimates (where relevant) and confidence intervals of the impacts of lock-down on measures of DV incidence and reporting under a range of identifying assumptions and specifications. In each case, standard panel event studies are displayed as hollow circles and 95% CIs, along with de Chaisemartin and D'Haultfœuille [2020]'s DID_M estimator with full dynamic lags and leads. For event studies, month -7 is used as the baseline omitted period, as this is pre-pandemic, even for municipalities which adopt lockdown in the final period of study. Specifications with epidemiological controls are indicated by hollow squares, and unconditional estimates are displayed as diamonds. Finally, Rambachan and Roth [2023]'s partially identified 'Honest DiD' bounds are presented as shaded purple areas, however now rather than assuming parallel counterfactual trends between lockdown and non-lockdown municipalities, project any prevailing trends forward, where these trends are additionally allowed to vary by as much as M between each period. Darker shaded areas correspond to $M = \{0.03, 0.005, 0.02\}$ for helpline calls, shelter residence, and crime reporting respectively, and lighter shaded areas correspond to $M = \{0.1, 0.03, 0.05\}$. In the case of shelter usage only, weekly, rather than daily estimates are plotted, given computational demands on bounds when many lags are included.

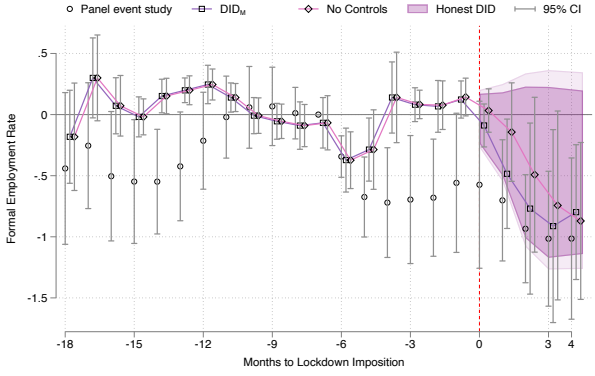
Figure A6: Estimated Impacts of Lockdown on Mechanism Variables



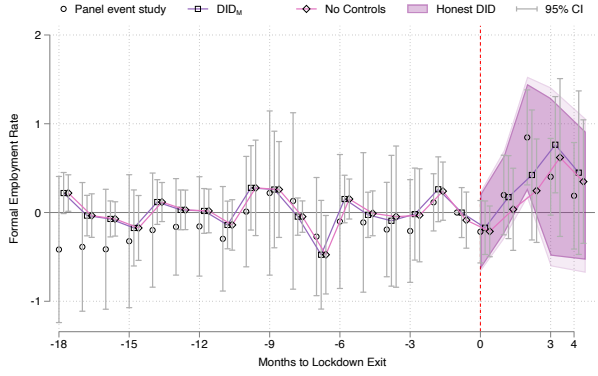
(a) Cell Phone Mobility Patterns (Entry)



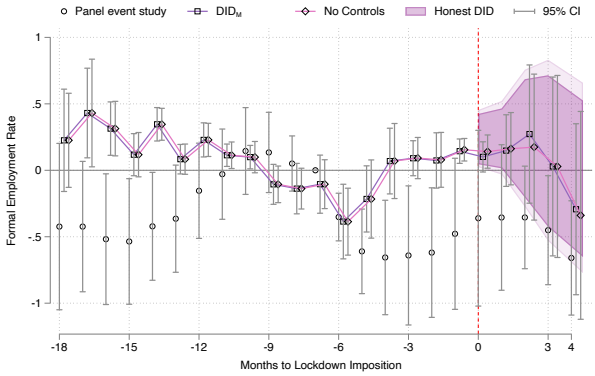
(b) Cell Phone Mobility Patterns (Exit)



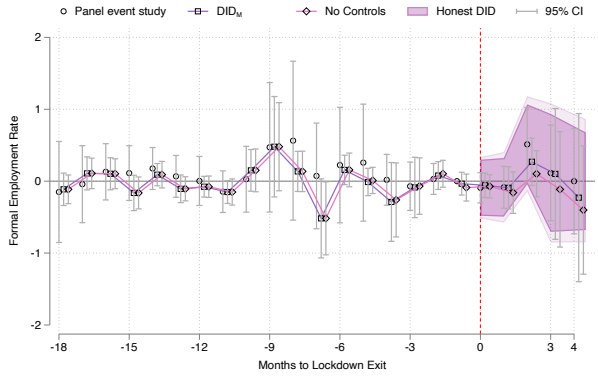
(c) Male Formal Employment Rate (Entry)



(d) Male Formal Employment Rate (Exit)



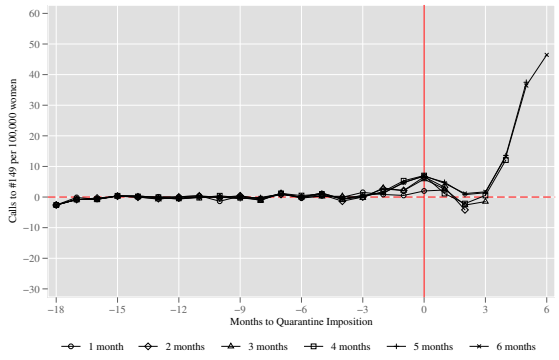
(e) Female Formal Employment Rate (Entry)



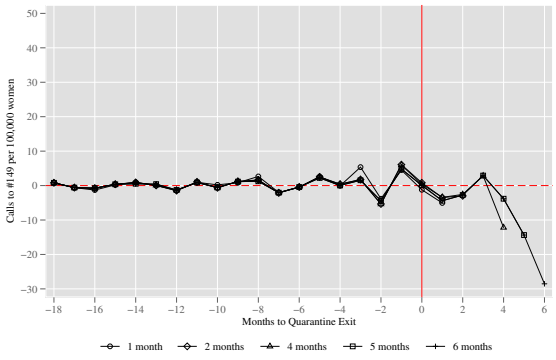
(f) Female Formal Employment Rate (Exit)

Notes: Refer to Notes to Figure A5. All details are identical, however here plots for each mechanism variable are displayed. In the case of Rambachan and Roth [2023]'s partially identified 'Honest DiD' bounds, darker shaded areas correspond to $M = \{0.05, 0.02, 0.02\}$ for mobility, male and female employment respectively, and lighter shaded areas correspond to $M = \{0.15, 0.10, 0.10\}$. In the case of shelter mobility only, weekly, rather than daily estimates are plotted, given computational demands on bounds when many lags are included.

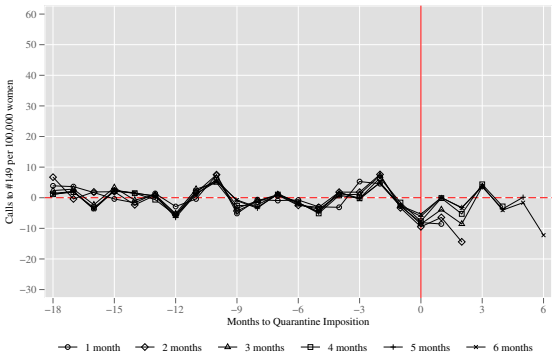
Figure A7: Heterogeneity in Lockdown Impacts by Lockdown Duration



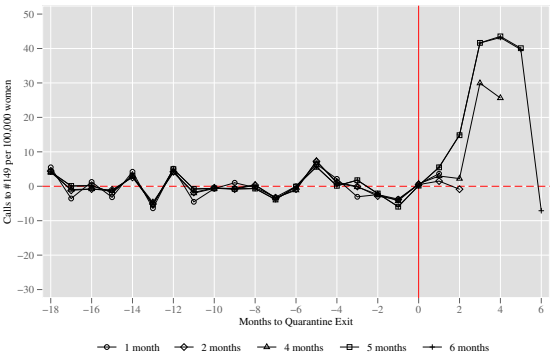
(a) Helpline Calls (Entry)



(b) Helpline Calls (Exit)



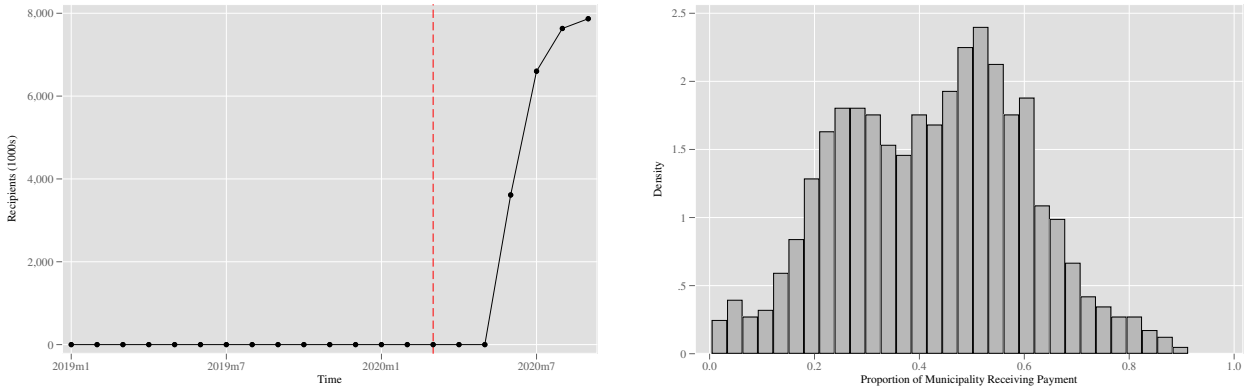
(c) Crime Reporting (Entry)



(d) Crime Reporting (Exit)

Notes: Refer to notes to Figure 2. Identical specifications are estimated however examining rates of calls to the Chilean police family support hotline (panels (a) and (b)), and formal criminal complaints (panels (c) and (d)) on different samples defined by maximum lockdown length: from 1 to 6 months (including municipalities with 0 months of lockdown). All other estimation details follow those indicated in notes to Figure 2.

Figure A8: IFE Stimulus Payments

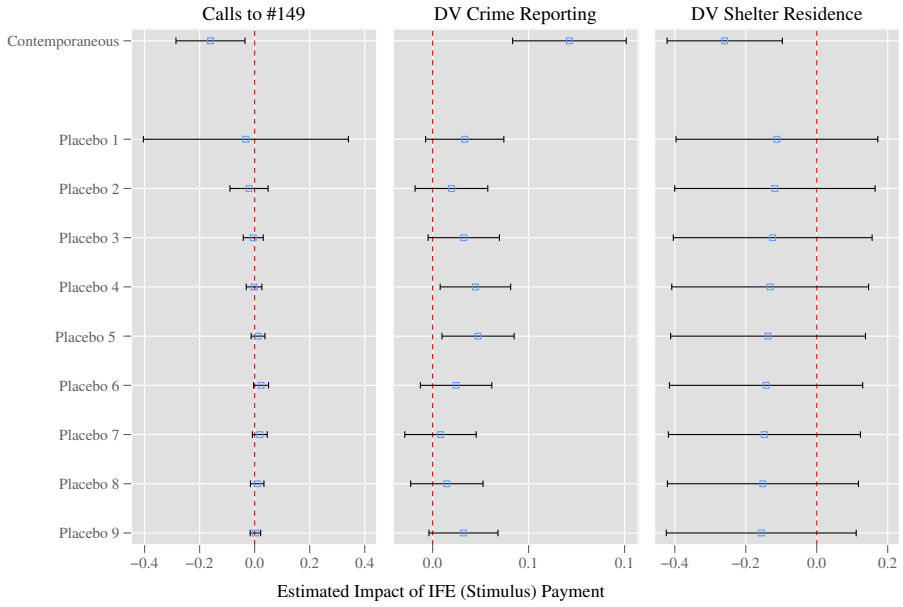


(a) Payments over Time

(b) Proportion of Recipients per Municipality

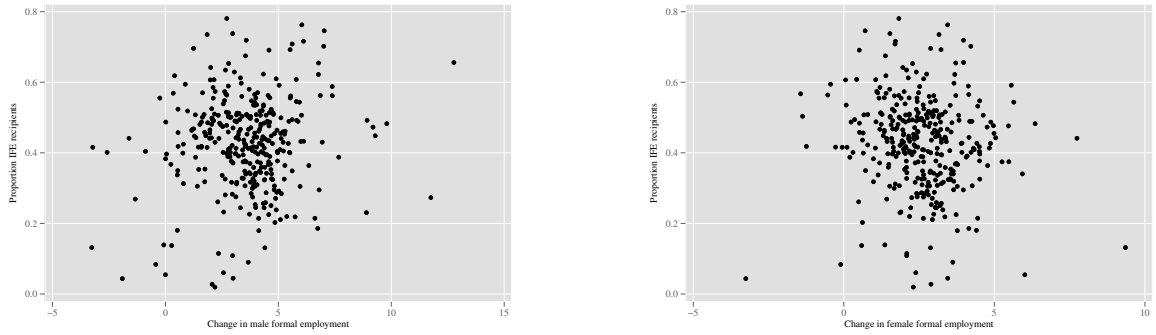
Notes: Panel (a) plots the temporal expansion of the number of IFE recipients nation-wide, based on administrative records provided by the Ministry of Social Development of Chile (Undersecretary for Social Evaluation). Panel (b) plots the average proportion of each municipality's residents receiving an IFE payment, calculated in the period following the introduction of the payment.

Figure A9: Placebo Tests – Stimulus Payment Impacts on DV Measures



Notes: Placebo tests consist of estimating two-way FE models with rates of IFE payments as the independent variable of interest on rates of domestic violence. In the case of 'Contemporaneous' models, this consists simply of a regression of rates of DV in time t on IFE (family emergency income payments) at time t, whereas placebos regress lagged rates of DV at time $t-k$ on IFE at time t. In the case of Calls and Crime, alternative placebos use lags $k \in \{10, 11, \ldots, 18\}$, such that in all cases DV outcomes are considered in months of 2019 or prior. In the case of shelters where daily data is used (starting from January 2020), lags $k \in \{60, 61, \ldots, 68\}$ are considered.

Figure A10: Stimulus Payment Coverage and Employment Losses



(a) Male Employment Change

(b) Female Employment Change

Notes: Plots for employment change and IFE recipients as share of population are displayed. Each dot is a municipality and values are calculated as averages for June-Sept 2020, which is the period when IFE was delivered. Employment change is defined as the average employment rate before COVID (Jan 2019-Feb 2020) minus the average employment rate after COVID (March-Sept 2020).

Table A1: Negative Weights in Two-way Fixed Effect Models

	Number g,t	Number Negative g,t	Total Sum of Negative Weights
Lockdown Entry			
Helpline Calls	334	15	-0.0084
Police Reporting	334	7	-0.0019
Shelters	1844	225	-0.0906
Lockdown Exit			
Helpline Calls	130	0	—
Police Reporting	130	0	—
Shelters	132	0	—

Notes: The number of group×time specific estimates are displayed, as well as the number of units receiving negative weights following de Chaisemartin and D'Haultfœuille [2020]. Each g,t group refers to treatment effects in municipalities which adopted at a specific time, in a specific post treatment year t. A full decomposition of the two-way fixed effect estimator for lockdown entry in models based on municipal data is provided in Figure A4.

Table A2: Proportion of Municipalities and Population Driving Estimates

	Lockdown Imposition		Lockdown Exit	
Time Period	Municipalities	Population	Municipalities	Population
0	128	16,568,433	43	5,808,802
1	91	14,076,074	43	5,808,802
2	59	10,831,426	18	1,978,723
3	41	7,571,212	9	694,585
4	27	5,094,053	9	694,585
5	7	1,707,023	–	–

Notes: DID_M dynamic (post-lockdown) effects are based on municipalities which have entered lockdowns, and been in lockdown for at least x months (or exited lockdowns, and been out of lockdowns for x months in the case of lockdown exit), where x refers to the time period indicated on the horizontal axis of the DID_M result graph. As lockdown length varies by municipality, estimates of varying length are driven by different sub-groups of municipalities. The sub-groups having had at least 0-6 months of lockdown are indicated in the left-hand panel, and the subgroups having had at least 0-6 months out of lockdown are indicated on the right-hand panel.

Table A3: DID_M Estimates of Lockdown Imposition and Removal on Helpline Calls by Type of Violence

	All Calls		Physical Violence		Psychological Violence		Economic Violence	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Lockdown Imposed	5.841***	5.012**	3.977***	2.305*	1.768	2.542**	0.096**	0.164***
	(1.888)	(2.273)	(1.143)	(1.200)	(1.110)	(1.222)	(0.039)	(0.058)
Lockdown Removed	-4.390**	-2.089	-2.383***	-1.214	-2.009	-0.907	0.002	0.032
	(1.791)	(1.686)	(0.806)	(1.255)	(1.441)	(1.240)	(0.042)	(0.054)
Baseline Mean	5.350	5.350	2.002	2.002	3.326	3.326	0.021	0.021
COVID-19 Controls		Y		Y		Y		Y

Notes: Each column displays aggregate DID_M estimates of impacts of lockdown imposition and removal on DV helpline calls by type of violence as classified by police telephone operators. Columns (1)-(2) replicate results using all calls (as in Figure 2), while columns 3-8 break this down by each violence type. Aggregate DID_M estimates are presented weighting by the number of affected individuals at each lag. Standard errors are estimated using a blocked bootstrap clustered by municipality. ***, ** and * indicate statistical significance at the 1%, 5% and 10%, respectively.

Table A4: Two-Way FE Models: DV and Mechanism Measures

	DV Measures			Mechanisms		
	Calls	Crime	Shelters	Mobility	Male Emp	Female Emp
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: No COVID Controls						
Lockdown Imposed	12.80***	-11.32***	0.0594	-0.567***	-0.397***	-0.00405
	(1.631)	(1.755)	(0.149)	(0.0925)	(0.113)	(0.0782)
Lockdown Removed	2.833	0.821	0.212	-0.0183	-0.162	-0.237
	(1.818)	(2.657)	(0.457)	(0.0853)	(0.201)	(0.171)
Observations	7,245	7,245	4,576	74,992	7,245	7,245
Panel B: COVID Controls						
Lockdown Imposed	8.627***	-6.072***	0.154	-0.599***	-0.318***	0.0639
	(1.268)	(1.737)	(0.166)	(0.0949)	(0.120)	(0.0922)
Lockdown Removed	1.722	-0.221	0.280	-0.0281	0.0738	-0.117
	(1.401)	(2.610)	(0.468)	(0.0824)	(0.209)	(0.177)
Diagnosed COVID Cases per 1,000 People	0.714***	-1.163***	-0.508	0.260**	-0.0243	0.00300
	(0.144)	(0.151)	(0.335)	(0.101)	(0.0160)	(0.0139)
PCR Tests per 1,000 People	-0.123***	0.0197	-0.0856	-0.0308	0.0262***	0.00643
	(0.0431)	(0.0614)	(0.145)	(0.0454)	(0.00548)	(0.00446)
Population (1,000s)	0.695***	-0.233			-0.0516***	-0.0401***
	(0.205)	(0.177)			(0.0164)	(0.0115)
Observations	7,245	7,245	4,528	74,992	7,245	7,245
Mean of Dep. Var. (baseline)	5.350	92.577	3.049	2.862	34.336	21.413
Mean of Indep. Var. (post-COVID)						
Diagnosed cases	2.614	2.614	0.109	0.086	2.614	2.614
PCR tests/1000	21.105	21.105	0.921	0.685	21.105	21.105

Notes: Each column displays a two-way fixed effect regression of the impact of lockdown on domestic violence measures (columns 1-3) or mechanism variables (columns 4-6). In columns 1-3, all measures are cast per 100,000 population. Column 4 refers to average trips within municipality per day, columns 5-6 are cast per 100,000 working age population. Lockdown Imposed is a binary variable taking the value of 1 only when municipalities are under lock-down, and 0 pre or post-lockdown. Lockdown removed switches to 1 post-lockdown in areas where lockdown has been imposed and then removed. Population is not included in columns 3-4 given that these are day by municipality measures for 2020 only. Fixed effects for area and time are consistently included, and standard areas are clustered by geographic unit. ***, ** and * indicate statistical significance at the 1%, 5% and 10%, respectively.

Table A5: Stimulus Payments, Lockdown and Domestic Violence (no COVID controls)

	Calls to 149		Crime Reporting		Shelter Residence	
	(1)	(2)	(3)	(4)	(5)	(6)
Lockdown	11.38***	14.76***	-10.12***	-10.40***	0.0682	0.155
	(1.462)	(2.341)	(1.851)	(2.410)	(0.134)	(0.153)
IFE	-9.359	-7.172	15.12***	14.94***	-3.081***	-3.021***
	(6.896)	(6.300)	(5.618)	(5.592)	(0.963)	(0.993)
Lockdown× IFE		-17.08**		1.410		-0.348
		(7.122)		(6.920)		(0.408)
IFE+						
Lockdown x IFE		-24.252*		16.351*		-3.369***
		(12.437)		(8.738)		(0.929)
Observations	5520	5520	5520	5520	4576	4576
R-squared	0.770	0.773	0.617	0.617	0.858	0.858

Notes: Each column regresses measures of DV incidence or reporting on time-varying receipt of IFE stimulus payments, which began in May, 2020, and further rolled out to broader populations thereafter. IFE refers to the proportion of the municipal (columns 1-4) or regional (columns 5-6) population receiving transfers. Lockdown×IFE interacts each municipality's lockdown status with the proportion of IFE receipt. All regressions include geographic and time fixed effects and cluster standard errors by area. "Joint Estimate" refers to the linear combination of the parameters on IFE and Lockdown×IFE along with the standard error of this combination, reported in columns where interactions are considered. *** p<0.01, ** p<0.05, * p<0.1.

B The Shift Share Design

In this Appendix, we set out the identifying assumptions of the shift share design used in Section 4.3 of the paper, and provide supporting evidence. Equation 2 of the paper describes how domestic violence depends on employment in a municipal by month panel. In Section 4.3 we explain that we address endogeneity of employment using the Bartik instrument. The estimated equations for male and female employment are:

$$DV_{st} = \alpha + \beta_1 \text{Male Employment}_{st} + \beta_2 \text{Female Employment}_{st} + \mu_s + \phi_t + X'_{st}\Gamma + \varepsilon_{st}, \quad (3)$$

The first stage equations of the IV model are:

$$\text{Female Employment}_{st} = \pi_0^F + \pi_1^F B_{st}^{female} + \pi_2^F B_{st}^{male} + \mu_s^F + \phi_t^F + X'_{st}\xi^F + \eta_{st}^F \quad (4)$$

$$\text{Male Employment}_{st} = \pi_0^M + \pi_1^M B_{st}^{female} + \pi_2^M B_{st}^{male} + \mu_s^M + \phi_t^M + X'_{st}\xi^M + \eta_{st}^M, \quad (5)$$

where B_{st}^G represents the Bartik instrument for gender $G \in \{female, male\}$, and is defined as in the body of the paper:

$$B_{st}^G = \sum_{k=1}^{K} z_{s0k}^G \times g_{kt}^G,$$

with k indexing employment sectors, G indexing sex (female or male), z are the baseline industry shares at the municipality level, and g is the aggregate COVID shock.

Although COVID is a single aggregate shock, it will tend to have different impacts on job loss in different municipalities on account of their different baseline industry shares. For example, municipalities in which a greater proportion of individuals at baseline were working in construction, the arts and tourism, where COVID had a larger impact will have suffered greater job loss, see Figure C3. This will vary by gender given systematic differences in baseline industry-employment shares by gender.

The shift share IV achieves identification through the conditional exogeneity of the *shares* (baseline exposure) as laid out in Goldsmith-Pinkham et al. [2020], or the *shocks* (COVID), as laid out in Borusyak et al. [2022]. In our setting, we argue that identification is drawn from the shares: at baseline certain municipalities were more exposed to the shock due to baseline sectoral structure, but the COVID shock need not be exogenous. In practice, this implies that consistency does not require the number of industries to be large, but rather the number of municipalities to be large. As we lay out when discussing Assumption 2 below, a fundamental point is not that these baseline shares are independent of municipal-level unobservables, but rather that they are independent of first differences in municipal-level unobservables.
A concern is that even conditional upon municipal and time fixed effects, employment changes may be correlated with unobservable time-varying factors. In particular, municipalities with (endogenously) higher COVID infection (or testing) rates may have had increased domestic violence through stress, or migration, conditional on job loss rates.

Identifying Assumptions We first set out the identifying assumptions, and then discuss the evidence to support them.

Assumption 1. *Relevance – Each shift share instrument must be correlated with employment, and the system must not be under-identified.*

Goldsmith-Pinkham et al. [2020] note that the lower level sufficient condition for this to hold is that there must be an industry and time period when the industry share has predictive power for employment, conditional on controls, and that the growth rates cannot weight covariances between industry share and employment such that they exactly cancel out. We will document the first stage relationships below, and provide details related to under-identification and weak instrument tests.

Assumption 2. *Strict exogeneity – Conditional on included controls and fixed effects, baseline industry shares must be uncorrelated with the structural error term in the following sense:*

$$E[\varepsilon_{st} z^G_{lk0} | D_{st}] = 0 \text{ for all } k \text{ and } G \text{ where } g^G_k \neq 0.$$

Here the notation D_{st} refers to observable controls and municipal and year FEs.

A key point here is that Assumption 2 does not imply that these shares are uncorrelated with municipal specific factors such as the skill distribution, rates of domestic violence at baseline, or other equilibrium conditions. This is because the assumption refers to exogeneity conditional on fixed effects and time-varying controls (refer to discussion in Goldsmith-Pinkham et al. [2020, p. 2598]). Municipal fixed effects will capture all time-invariant factors, and as such, we require

baseline shift shares to be uncorrelated with *changes* in the error term. Identification in this setting requires that the differential effect of higher exposure to a more COVID-impacted industry only impacts domestic violence through employment, and not through other (confounding) time-invariant channels.

The main potential concern in our setting relates to COVID-transmission. It is plausible that a higher concentrations of baseline industries such as construction or hospitality that were more vulnerable to COVID transmission could, through higher COVID infection rates, lead to more stress and, as a result more DV, violating the exclusion restriction. To address this concern, we consistently control for COVID transmission and testing as X_{st} in equation 2. This increases our faith in the conditional exogeneity assumption. It seems unlikely that there are other relevant time-varying factors correlated with baseline sector shares, especially in the relatively short time period of our analysis. Most relevant factors are likely sufficiently slow-moving that they will be captured by municipal fixed effects. We nevertheless proceed to conduct empirical tests designed to examine the plausibility of the identifying assumptions, following Goldsmith-Pinkham et al. [2020].

Testing the Plausibility of The Identifying Assumptions We can test for relevance and under-identification formally, by examining first stage regressions of employment on the shift share instrument. These are presented in Table B1. Here we observe clearly that the instruments for male and female employment are relevant and strong. As expected, baseline employment shares for women are most predictive for employment for women, and baseline employment shares for men are most relevant for explaining employment of men. As we have two instruments (the Bartik for men and for women), we are able to reject a null of under-identification based on the Sanderson and Windmeijer [2016] procedure. We are also able to reject the null of weak IV, based on the Cragg-Donald F-statistic reported in Table 1.

Table B1: Shift Share First Stage

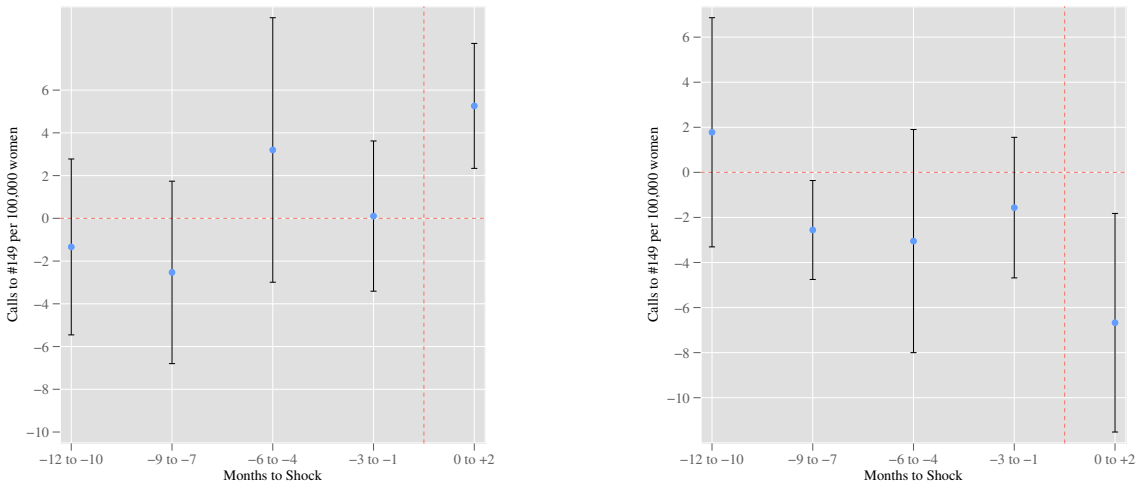
	Formal Employment	
	Women	Men
	(1)	(2)
Bartik Women	0.111***	0.000250
	(0.00872)	(0.00918)
Bartik Men	-0.0192	0.0763***
	(0.0167)	(0.0198)
Observations	5,520	5,520
Joint F-Test of First-stage	98.054	10.871
Weak IV (p-value)	<0.01	<0.01
Under-identification test F-statistic	163.044	78.601
Under-identified (p-value)	<0.01	<0.01

Notes: First stage estimates are reported on each endogenous variable. *** p<0.01, ** p<0.05, * p<0.1.

While we cannot formally test the strict exogeneity assumption given that it is an assumption

about unobservable terms, we can follow the literature in seeking to provide partial support for the strict exogeneity assumption by considering parallel pre-trends. We follow [Goldsmith-Pinkham et al., 2020, §VII.E] in implementing these tests. The logic of a parallel pre-trend test is that if the shift share instrument leverages identification based on a shock which is channelled through conditionally exogenous industry shares, then the IV should have no impact on the outcome of interest before the shock occurs. Figure B1 presents results from a series of regressions where we estimate the Bartik specification, but for bunched time periods before and after the shock. In each case, the outcome is the number of helpline calls per 100,000 women. For female and male employment, the figures make clear that there are no pre-trends, but that a significant impact of job loss on helpline calls emerges after the COVID shock, the timing of which is defined to be zero in event time.

Figure B1: Test of Parallel Pre-Trends in Shift Share Approach



(a) Female Job Loss (Bartik lags)

(b) Male Job Loss (Bartik lags)

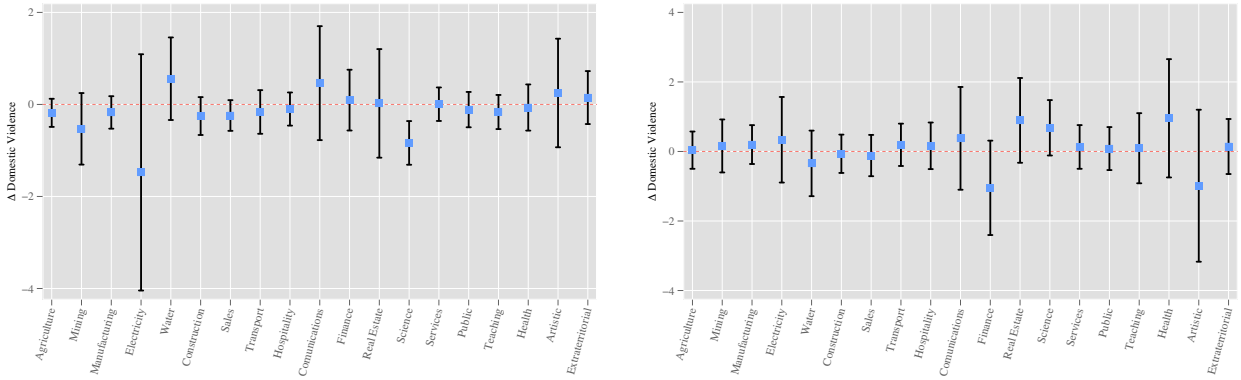
As an additional partial test of the assumption that baseline (pre-COVID) industry shares are exogenous in the 2SLS model using the Bartik instrument, we test whether these baseline shares are correlated with changes in domestic violence in the pre-COVID period. The Bartik IV rests on the premise that the baseline shares determine the intensity of exposure to an aggregate COVID shock which causes local DV. If this is right, we expect a correlation between industry employment shares and changes in DV after the shock but not in the pre-shock period. Figure B2 shows that the data satisfy this condition. Across 38 different industry shares, only one significant effect is observed.

Are There Interactive Effects Between Male and Female Job Loss? The main analysis, reflected in Equation 3, allows that male and female employment influences rates of DV, but assumes that these effects are independent of the level of each other. We now investigate a more general specification that allows multiplicative effects. Specifically, we estimate:

$$DV_{st} = \alpha + \beta_1 \text{Male Emp}_{st} + \beta_2 \text{Female Emp}_{st} + \beta_3 (\text{Male Emp} \times \text{Female Emp})_{st} + \mu_s + \phi_t + X'_{st}\Gamma + \varepsilon_{st},$$

We estimate this model using the shift share approach as before, with each of the 3 employment

Figure B2: Correlations Between Baseline (Pre-Covid) Industry Shares and Pre-Covid Δ Domestic Violence



(a) Women's Industry Shares

(b) Men's Industry Shares

Notes: Each plot presents point estimates and 95% confidence intervals from a regression of first differences (calculated by differencing month by month) in domestic violence (DV helpline calls) on municipal level employment shares for each municipality in 2019 (pre COVID-shock). A single regression is estimated in each sub-plot. Standard errors are clustered by municipality.

terms instrumented by their respective shift share IV. The results are in Table B2. When the outcome is crime reporting, there is no interaction between male and female employment. When the outcome is DV calls, there is a meaningful but small interaction. The increase in calls following male job loss is larger when women are also losing their jobs. This is consistent with our interpretation that the mechanisms triggered by male job loss are income-related stress and exposure. Looking at how the interaction modifies the decrease in calls following female job loss, we see that it is attenuated when men are also losing their jobs. This is consistent with our interpretation that the mechanism triggered by female job loss is that she becomes more tolerant of being victimized (her participation constraint is relaxed).

Table B2: Testing for Additional Interactive Employment Effects

	Calls				Crime
	All Calls	Economic	Physical	Psychological	Reporting
Formal Employment (Women)	3.017***	0.00155	1.680***	1.336***	2.948**
	(0.801)	(0.0195)	(0.472)	(0.395)	(1.187)
Formal Employment (Men)	-2.378***	0.00568	-1.447***	-0.936***	-1.171
	(0.707)	(0.0234)	(0.449)	(0.359)	(1.442)
Interaction (Women × Men)	-0.0249***	-0.000147**	-0.0124***	-0.0124***	-0.00191
	(0.00516)	(0.0000597)	(0.00251)	(0.00275)	(0.00476)
Observations	5,520	5,520	5,520	5,520	5,520
Mean Dep. Var.	5.311	0.020	1.970	3.321	91.169
First Stage F-Statistic (C.-D.)	303.3	303.3	303.3	303.3	303.3

Notes: See Table 1 in the main text. These are results for panel A generalized to allow that impacts of a change in the male employment rate depend upon the female employment rate, and vice versa.

C Data Appendix

Table C1: Data Definitions and Sources

Variable	Description	Frequency	Unit of analysis	Period covered	Source
Helpline calls to the Police DV hotline	Number of calls received by the domestic violence police line #149 made by women per 100,000 women. The calls are classified as owing to complaints related to physical violence, psychological violence, and economic violence.	Monthly	Municipality	Jan 2019 - Sept 2020	Carabineros de Chile
Crimes reported for DV	Domestic violence crimes known to the police with a woman victim. They include formal complaints made to the police, as well as in flagrante offences	Monthly	Municipality	Jan 2015 - Sept 2020	Ministry of Interior, Undersecretary of Crime Prevention
Residents in DV Shelters	Number of residents in government-run domestic violence shelters measured as the number of women who slept in a shelter the night before per 100,000 women	Daily	Region	Jan 1, 2020 - Nov 6, 2020	Ministry of Women and Gender Equality
Spots in DV Shelters	Number of official spots available in government-run domestic violence shelters per 100,000 women	Daily	Region	Jan 1, 2020 - Nov 6, 2020	Ministry of Women and Gender Equality
Lockdown entry	Dummy indicating a municipality is under lockdown	Daily	Municipality	Mar 14, 2020 - Sept 30, 2020	Ministry of Health, hand compiled by project RA.
Lockdown exit	Dummy indicating a municipality has exited lockdown	Daily	Municipality	Mar 14, 2020 - Sept 30, 2020	Ministry of Health, hand compiled by project RA.
COVID-19 testing rate	Number of COVID-19 PCRs over 1,000 inhabitants	Daily	Municipality	Mar 8, 2020 - Sept 30, 2020	Ministry of Science
COVID-19 infection rate	Number of confirmed COVID-19 cases over 1,000 inhabitants (7-day average)	Daily	Municipality	Mar 8, 2020 - Sept 30, 2020	Ministry of Science
Mobility	Number of trips within municipality measured by changes in cell phone connection towers	Daily	Municipality	Feb 26, 2020 - Sept 30, 2020	Cell phone metadata [Bravo and Ferres, 2020].
Employment	Number of employed individuals in the formal private sector over working age population	Monthly	Municipality	Jan 2019 - Sept 2020	Unemployment Insurance data managed by the Pensions Superintendence
Mean salary	Mean salary among employed individuals in the formal private sector	Monthly	Municipality	Jan 2019 - Sept 2020	Unemployment Insurance data managed by the Pensions Superintendence
Contract suspension rate	Number of jobs furloughed under the employment protection law over 100,000 pre-COVID jobs	Monthly	Municipality	April 2020 - Sept 2020	Pensions Superintendence
Jobs sectoral shares	Share of private sector workers under contract working in different economic sectors	Monthly	Municipality	Jan 2012 - Sept 2020	12% random sample of affiliated workers to the Unemployment Insurance, data managed by the Pensions Superintendence
Stimulus payment	Number of people receiving the IFE benefit over population		Municipality	May 2020 - Sept 2020	Ministry of Social Development and Family

We use three measures of domestic violence which capture incidence and reporting: calls received by the police domestic violence hotline, criminal cases of domestic violence, and use of shelters for victims of domestic violence. The data cover the whole country, with information either at the municipal or regional level.

Figure C1a plots the national trend for calls received by the police to the #149 domestic violence hotline. The figure shows a sharp increase of around 300% following the first outbreak of COVID-19, with the first lockdown measures soon after. Prior to March 2020 around 1,000-1,500 calls per month were received, with this quantity increasing to anywhere between around 2,000-5,000 calls per month after March 2020.

Figure C1b shows the national trend for formal criminal cases of domestic violence filed with the police. In this case, we observe daily data from January 1 2018 to 31 May 2020. We see that by mid-March 2020, the rate of DV criminal complaints differs from that of previous years, in particular, being lower than the rate observed in 2018 and 2019.

In the case of shelters, we observe data for each one of the publicly run women's shelters in the country, recording their number of residents, and available spots on each day between January 1, 2020 and November 6, 2020.[12] Not all municipalities have a shelter. In emergency situations, individuals are placed in a shelter in their region (which contains their municipality). The daily averages for rate of residence in DV shelters and total capacity for the period under study are summarized in Table C2, and national trends are plotted in Figures C1c and C1d, respectively.

We additionally have information from late February 2020 (before cases of COVID-19 were detected in Chile) recording movement within and between municipalities based on cellphone records. Baseline mobility within municipalities is approximately 3 trips per day (see Table C2), with trips measured as the average number of changes between cell phone connection towers from cell phone metadata.[13] Figure C2a displays mobility rates, showing a significant decrease in mobility since the start of the pandemic, that is more pronounced in municipalities subject to lockdown.

We have labor market data containing information on individual monthly work histories, along with sector of work, and whether workers suffered job loss. Figure C3 shows sectoral trends in formal employment for men and women, with the red line marking the first outbreak of the pandemic. We observe particularly sharp declines in construction, hospitality and artistic activities after March 2020. The construction sector employs a large share of the labor force, and mostly men. We supplement this information with data from a rolling labor market survey. Recognizing that the survey is not designed to be representative at the municipality level, we do not rely upon these estimates but provide them for descriptive purposes as they cover the entire population of formal and informal sector workers. Figure C1 shows increases in unemployment and declines in labor force participation rates for both men and women following the COVID-19 outbreak in March 2020. They reveal that, starting from a lower baseline level, unemployment rates of men increased more than rates for women. Starting from a lower base too, labor force participation rates of women fell more than rates for men.

Finally, we obtained information on the total beneficiaries of the stimulus payments distributed since May 2020. This information comes from the *Registro Social de Hogares*.[14] These payments

[12]For this variable we extended the period beyond September 30 in order to use all the administrative information available. This allows us to pick up more lockdown exits in our data and to have more variation for a variable that is defined at the regional level.

[13]See [Pappalardo et al., 2020] who also provide a description of the data.

[14]This household register, (RSH for its initials in Spanish), is an information system used by the Ministry

were targeted at those families most severely affected by the socioeconomic crisis brought on by the pandemic. Figure A8 shows the scope of stimulus payments up to September 2020. During this time, four payments were distributed.

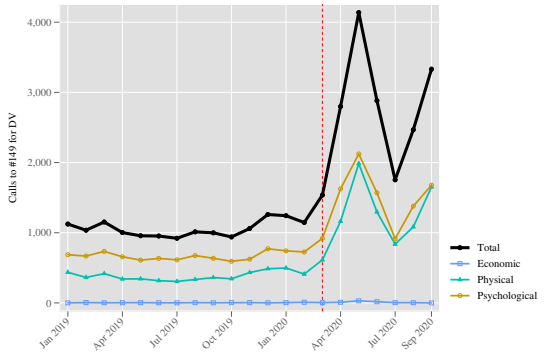
Table C2: Summary Statistics of Principal Measures

	Obs	Mean	Std. Dev.	Min.	Max.
Panel A: Helpline Calls					
All Calls to #149 per 100,000 Women	7245	8.15	14.99	0.0	325.7
Calls to #149 for Economic Violence per 100,000	7245	0.03	0.65	0.0	45.7
Calls to #149 for Physical Violence per 100,000	7245	3.33	9.24	0.0	325.7
Calls to #149 for Psychological Violence per 100,000	7245	4.78	9.05	0.0	200.0
Panel B: Criminal Complaints					
Complaints to Police for DV per 100,000 Women	7245	90.35	75.10	0.0	2985.1
Complaints to Police for DV per 100,000 Men	7245	22.77	24.70	0.0	392.2
Panel C: Women's Shelters					
Residents in DV Shelters per 100,000 Women	4576	3.21	2.43	0.0	13.7
Total Capacity of DV Shelters per 100,000 Women	4576	6.37	3.51	0.0	14.8
Panel D: Lockdown and Mobility Measures					
Quarantine Imposition	75428	0.11	0.32	0.0	1.0
Quarantine Exit	75428	0.04	0.21	0.0	1.0
Mobility within the Municipality per day	74992	2.33	1.82	0.0	16.1
Panel E: Covid Measures					
Diagnosed COVID cases per 1,000 people	7245	0.87	2.79	0.0	75.3
PCR tests per 1,000 people	7245	7.03	16.38	0.0	339.0
Panel F: Economic Measures					
Estimated Monthly Employment Rate (Women)	7245	20.53	9.55	0.0	147.1
Estimated Monthly Employment Rate (Men)	7245	33.21	11.99	0.0	122.5

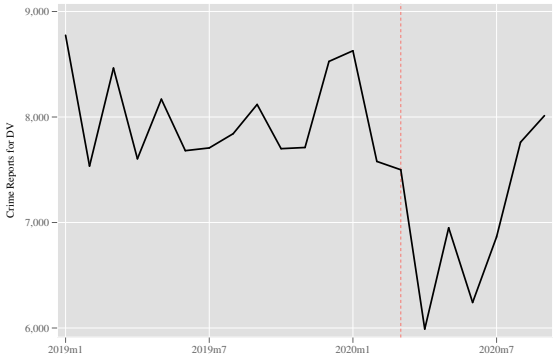
Notes: Summary statistics cover measures used in the paper at the same level used in analysis. For panels A, B, E and F this is by municipality and month. For panel C this is by region (which includes multiple municipalities) and day. For panel D, this is municipality by day.

of Social Development to assign a wide set of subsidies and social programs. It combines verified self-reported information with administrative records.

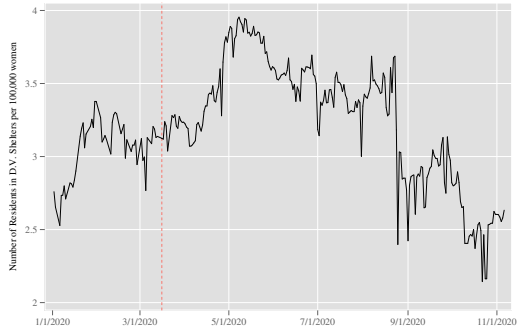
Figure C1: Descriptive Trends – Main Outcomes



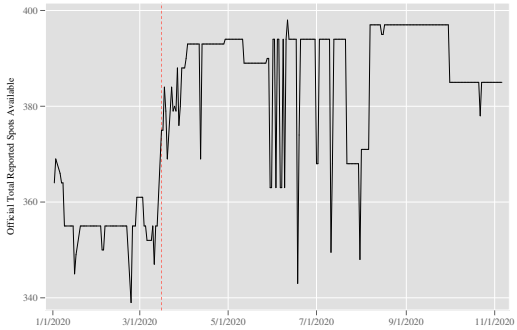
(a) Calls to #149



(b) Criminal Complaints for VIF



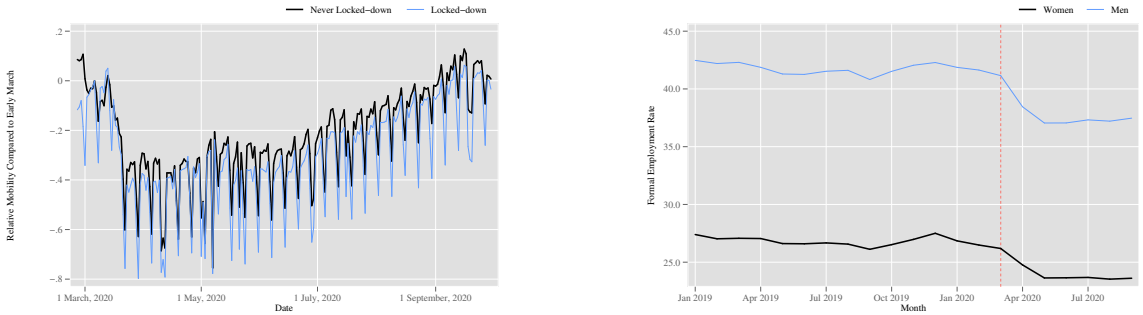
(c) Residents in Domestic Violence Shelters



(d) Spots in Domestic Violence Shelters

Notes: Panel (a) shows total aggregate number of calls in the country for each month between Jan 2019 - Sept 2020. The thick black line shows the total number of all calls related to DV received by the police. These are then sub-classified as related to psychological, physical and economic violence. Vertical red line indicates March 2020, the first month in which a lockdown was applied. Panel (b) shows rates of crime reporting for DV. These are plotted based on the number of reports received nation-wide each day. Grey lines depict rates for the first 5 months of years 2018 and 2019, while the red line depicts similar rates for 2020. In all analyses in the paper, month by municipality aggregates (over a longer period of time) are used. Panel (c) shows the average rate of residence in government run DV shelters, calculated as the total proportion of residents who slept in a shelter the previous night. This figure covers state-run women's shelters only. Panel (d) documents the total number of available spots in these state-run DV shelters according to administrative records.

Figure C2: Descriptive Trends – Mechanism Variables

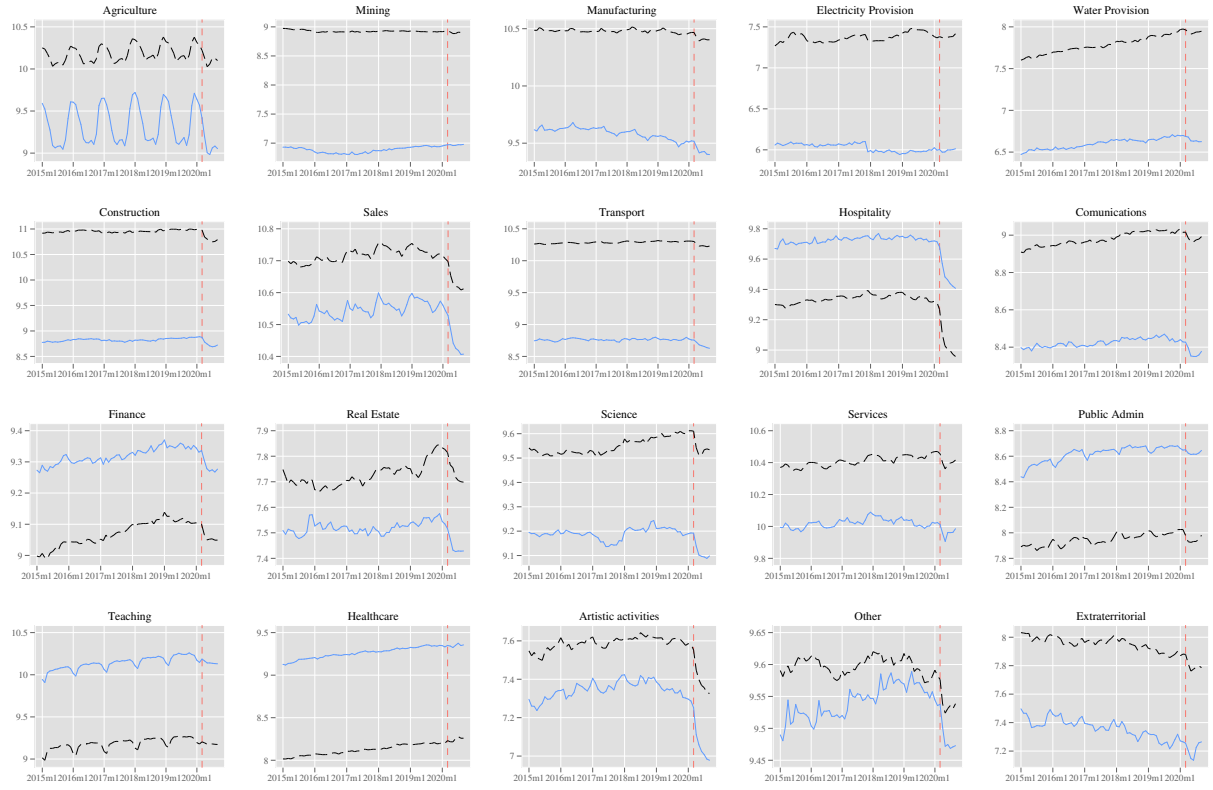


(a) Changes in Mobility by Lockdown Status

(b) Formal Employment Rates

Notes: Panel (a) presents daily changes in mobility (relative to early March) for municipalities that were locked-down compared to those that were never locked-down. Panel (b) presents monthly formal employment rates for men and women. The data for (b) comes from the 12% sample of unemployment insurance records covering the entire formal labour market.

Figure C3: Sectoral Trends in Employment by Gender (Total Employment)



Notes: Data from a 12% sample of the universe of formal private workers is displayed as reported in the country's unemployment insurance database. The log number of workers in each employment class is displayed. Dashed black lines represent men's emplyment, and solid blue lines represent women's employment. Plots showing absolute values are available in Bhalotra et al. [2021a].